

Glyphosate induces growth of estrogen receptor alpha positive cholangiocarcinoma cells via non-genomic estrogen receptor/ERK1/2 signaling pathway

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ABSTRACT

Previous studies showed that glyphosate stimulates breast cancer cell growth via estrogen receptors. The present study investigated the effect of glyphosate on the estrogen signaling pathway involved in the induction of cholangiocarcinoma (CCA) cell growth. HuCCA-1, RMCCA-1 and MMNK-1 were chosen for comparison. The effects of glyphosate on cell growth, cell cycle and molecular signaling pathways were measured. The results showed that HuCCA-1 cells expressed estrogen receptor alpha (ER α), while ER α was not detected in RMCCA-1 and MMNK-1 cells. ER α was mostly expressed in cytoplasmic compartment of HuCCA-1 cells. Estradiol (E2) (10^{-11} – 10^{-5} M) induced cell proliferation in HuCCA-1 but not in RMCCA-1 and MMNK-1 cells. Glyphosate at the same concentration range also induced HuCCA-1 cell proliferation. The S phase of the cell cycle, and protein levels of the cyclin family were significantly increased after treatment of glyphosate or E2. Both compounds also induced the expression of proliferative signaling-related proteins including ER α , VEGFR2, pERK, PI3K(p85), and PCNA. These effects of glyphosate and E2 were abolished by the ER antagonist, 4-hydroxytamoxifen and U0126, a MEK inhibitor. The data from this study indicate that glyphosate can induce cell growth in ER α positive CCA cells through non-genomic estrogen receptor/ERK1/2 signaling pathway.

1. Introduction

Glyphosate [N-(phosphonomethyl)glycine] is the main active ingredient in one of the most widely used herbicides in the world. The commercially used concentrations of glyphosate range from 1% for domestic use to 41% for conventional agricultural uses (Bradberry et al., 2004). Glyphosate has a specific mode of action: the inhibition of enzyme, 5-enolpyruvylshikimate acid-3-phosphate synthase (EPSP) (Steinrücken and Amrhein, 1980), an enzyme that is present only in plants and some microorganisms. In Thailand, glyphosate is the highest imported herbicide commonly used in agriculture (Apiwat et al., 2014). Recently, glyphosate and its major metabolite, aminomethylphosphonic acid (AMPA), have been detected as contaminants in the environment, food chain and agricultural products (Bai and Ogbourne, 2016; Rubio et al., 2014). Interestingly, the presence of glyphosate and AMPA in human serum and breast milk has recently been reported (Ehling and Reddy, 2015; Kongtip et al., 2017; Steinborn et al., 2016), raising

concerns about human exposure to these chemicals. Some studies suggest that glyphosate has no or low toxicity in animals and humans; however, other studies have indicated that they can induce toxic effects. For examples, studies have revealed that glyphosate can inhibit the function of cytochrome P450 (CYP450), an important superfamily of metabolizing enzymes (Hietanen et al., 1983; Samsel and Seneff, 2013). Other studies have demonstrated the toxicity of glyphosate to several systems in the rat, including the renal and nervous systems (Hernandez-Plata et al., 2015; Karimi et al., 2013). An epidemiological study on cancer incidence in pesticide applicators showed a statistically significant relationship between the increased risk for certain cancers, such as multiple myeloma, and levels of glyphosate usage (De Roos et al., 2005). Several case-control studies from the USA, Canada and Sweden have indicated an increased risk for non-Hodgkin's lymphoma from occupational exposure to glyphosate after adjustment for other pesticide exposures (Guyton et al., 2015). Interestingly, a long-term study shows that rats exposed to glyphosate-based herbicide and/or

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glyphosate-tolerant maize over 2 years exhibited a wide range of endocrine disruptive effects, including modification of sex hormones and the development of large mammary tumors in female rats (Seralini et al., 2014).

Cholangiocarcinoma (CCA) is a malignant tumor arising from the epithelial cells of the biliary system. The increasing incidence and mortality of CCA have been reported worldwide. North-East region of Thailand has the world's highest chronic infection incidences related to the liver fluke, *Opisthorchis viverrini*, which is different from what found in Western countries where the primary sclerosing cholangitis (PSC) is a major related factor. However, there are some possible risk factors that could be considered as other important contributors to CCA (Banales et al., 2016). A previous study suggested that environmental toxicants, such as nitrosamines, may be a major factor that can contribute to the occurrence of CCA (Patel, 2011). It has been shown that glyphosate can be excreted into the bile, suggesting a potential exposure to cells in the biliary system (JMPR, 2004). One of the possible cancer promotion pathways of glyphosate is the disruption of the estrogen function (E2), and estrogen receptor (ER) signaling pathway. It has been suggested that glyphosate can act as an endocrine disruptor (Mnif et al., 2011). In 2005, Richard and coworkers reported that the glyphosate-based herbicide disrupts the aromatase activity and mRNA levels of the enzyme aromatase which is involved in steroid hormone synthesis (Richard et al., 2005). *In vitro* experiments have shown the exposure to glyphosate resulted in the proliferation of human breast cancer cells via the estrogen receptor (Thongprakaisang et al., 2013). The induction of breast cancer cell growth by glyphosate was also supported by the recent study of Mesnage and coworkers (Mesnage et al., 2017). However, the effects reported by this group were observed at relatively high concentrations of glyphosate.

The proliferation of CCA cell lines can be stimulated by estrogen and other growth factors (Alvaro et al., 2006; DeMorrow, 2009; Mancino et al., 2009). It has also been reported that estrogen positively modulates cholangiocyte proliferation, which is associated with cholangiopathies found in humans (Alvaro et al., 2004). However, the effects of glyphosate on CCA development are limited. This study aims to evaluate the effects of glyphosate compared to E2 in CCA cells. The range of concentrations that were used in this study is comparable to levels that have been reported in environmental human exposures (Acquavella et al., 2004; Bai and Ogbourne, 2016; Kongtip et al., 2017). Cell proliferation, effects on the cell cycle, and mRNA and protein expressions of various signaling proteins were analyzed. 4-hydroxytamoxifen, an ER antagonist, and U0126, an MEK inhibitor, were also used to determine effects of glyphosate exposure on the involving signaling pathways.

2. Materials and methods

2.1. Chemicals and cell lines

Glyphosate (> 98%) was purchased from AccuStandard (New Haven, USA). 17 β -estradiol (E2) was obtained from Sigma-Aldrich (Sigma-Aldrich, USA). The tamoxifen metabolite, (Z)-4-hydroxytamoxifen, an ER antagonist, was provided by Tocris Bioscience (Tocris Bioscience, UK). The MEK1/2 inhibitor, U0126, was purchased from Cell signaling Technology Inc (USA). The human intrahepatic CCA cell lines, HuCCA-1 and RMCCA-1, derived from bile duct tumor mass from Thai CCA patients, were established and kindly provided by Professor Stitaya Sirisinha (Sirisinha et al., 1991), and Dr. Kawin Leelawat (Rattanasinganchan et al., 2006), respectively. The immortalized cholangiocyte cell line, MMNK-1, which represented normal cholangiocytes, was ordered from Japanese Collection of Research Bioresources (JCRB) Cell Bank (Osaka, Japan). The MCF-7 breast cancer cell line, which was used as a positive model for the estrogen receptor dependent pathway were purchased from the American Type Culture Collection (ATCC, USA).

2.2. Cell culture conditions

Both HuCCA-1 and RMCCA-1 cells were maintained in HAM's F-12 medium supplemented with 10% fetal bovine serum (FBS) (JR Scientific Inc., USA), 2 mM L-glutamine, 100 units/mL penicillin and 100 μ g/mL streptomycin (Gibco, USA), at 37 °C in a 5% CO₂ humidified atmosphere. MMNK-1 cells were maintained in DMEM medium supplemented with 10% FBS, 2 mM L-glutamine, 100 units/mL penicillin and 100 μ g/mL streptomycin (Gibco, USA), at 37 °C in a 5% CO₂ humidified atmosphere. MCF-7 cells were maintained in MEM media supplemented with 10% FBS, 2 mM L-glutamine, 100 units/mL penicillin and 100 μ g/mL streptomycin, 1% v/v non-essential amino acids, 1 mM sodium pyruvate and 10 mg/L insulin, at 37 °C in a 5% CO₂ humidified atmosphere. After the confluence of the cells (approximately 80%), cells were sub-cultured. The culture plates were then maintained in a 5% CO₂ humidified incubator at 37 °C.

2.3. Cell culture and treatment conditions

For the estrogen withdrawal condition, cells were cultured in 100 mm sterile plates and allowed to reached approximately 60% confluence. Cells were then washed twice with sterile PBS. Next, cells were cultured in 10% dextran-charcoal stripped FBS (CSS) (HyClone, USA) in a non-phenol red RPMI medium (Sigma, USA) containing all supplementations except FBS for 4 days prior to seeding. After 24 h incubation for cell attachment, the cells were treated with various concentrations of test compounds and positive controls; 17 β -estradiol (E2), in non-phenol red RPMI medium containing 5% CSS with all supplements according to our previous study (Thongprakaisang et al., 2013). For serum free treatment conditions, after seeding and incubation for 24 h for cell attachment, cells were cultured in serum free medium (SF) 24 h prior to treatment.

2.4. Cell viability

PrestoBlue™ reagent is quickly reduced by metabolically active cells, providing a quantitative measure of viability and cytotoxicity. Before cell seeding for the PrestoBlue™ assay, cells were cultured in estrogen-withdrawal conditions. After treatment, 10 μ L of PrestoBlue™ reagent (Invitrogen Corp., USA) was added into each well and incubated at 37 °C for 30 min. The fluorescence was measured at an excitation/emission wavelength of 560/590 nm using a SpectroMax M3 microplate reader (Molecular Devices, USA), and expressed as the percentage of cell viability compared to controls. The remaining media was then removed and 100 μ L of 500 μ g/mL of MTT-containing medium was added to each well. Cells were incubated for 4 h, and then lysed with dimethyl sulfoxide (Sigma-Aldrich, USA). The optical density (OD) was measured at 570 nm and a reference wavelength of 650 nm using a SpectroMax M3 microplate reader, and was expressed as the percentage of cell viability compared to controls.

2.5. Cell cycle analysis

After maintaining in estrogen-withdrawal conditions, CCA cells were seeded into 60 mm plates (1.5×10^6 cells/plate) and cultured overnight. The cells were then treated with test compounds and cultured for another 48 h. The cells were trypsinized with phenol red-free trypsin (Gibco, USA) and washed with cold phosphate buffer saline (PBS). Subsequently, cells were fixed with cold 70% ethanol and incubated at 4 °C for 1 h and then washed with PBS. Cells were then stained by adding 1 mL of 50 g/mL propidium iodide (PI) solution (Sigma-Aldrich, USA) and 0.5 ng/mL RNase (Sigma-Aldrich, USA). Analysis was performed with a BD FACSCanto flow cytometer (BD Biosciences, USA) and cell cycle distributions were analyzed by the ModFit LT software (Verity House Software, USA).

2.6. mRNA expression analysis

Quantitative real-time RT-PCR (qRT-PCR) was used to determine mRNA expression of several genes, including *ERα*, *ERβ1*, *ERβ2*, *ps2*, *progesterone receptor (PR)* and *β-actin*. Total RNA was extracted using the RNeasy mini kit (QIAGEN, Germany) according to instructions by the manufacturer. The remaining residual DNA in RNA samples was removed with RNase-Free DNase (QIAGEN, Germany). Quantitative real-time PCR was performed using RNA-direct SYBR Green Realtime PCR Master Mix (Toyobo, Japan) in an Applied Biosystems StepOnePlus Real-time PCR (Life Technologies, USA). The *ERα*, *ERβ1*, *ERβ2*, *ps2* and *progesterone receptor (PR)* primers have previously been described (Nakareangrit et al., 2016). The *β-actin* primers are (forward strand): 5'-GCCTTTGCCGATCCGC-3' and (reverse strand): 5'-GCCGTA GCGGTGTGTCG-3'. The PCR conditions were as follows; reverse transcription (RT activation at 90 °C, 30 s, RT at 61 °C, 20 min, RT deactivation at 95 °C, 1 min); followed by 40 cycles of real-time PCR (15 s of denaturing at 95 °C, 15 s of annealing at 60 °C, 30 s and extension at 72 °C, 30 s). Relative mRNA expression was calculated after normalization with the level of a reference gene (*β-actin*) according to the $2^{-\Delta\Delta CT}$ method (Rao et al., 2013).

2.7. Western blot analysis

Cell lysates were prepared by placing cells in lysis buffer containing 10 mM Tris (pH7.4), 150 mM NaCl, 1% Triton X-100, 1 mM PMSF, 1 mM Na_2VO_4 , 20 mM NaF and 1× protease inhibitor cocktail set I (Calbiochem, USA). Then cell lysates were sonicated for 30 min at 4 °C before centrifugation at 14,000 rpm for 15 min at 4 °C. The protein concentrations were determined by the Bradford reagent (Bio-rad, USA). Equal amounts of total protein (50 µg) from each sample were loaded on 7.5% SDS-polyacrylamide gels and electrophoretically transferred to nitrocellulose membranes (Pall Corporation, USA). The membranes were blocked in 5% non-fat milk in Tris-buffered saline (TBS) containing 0.1% Tween-20 for 1 h at room temperature and probed with primary antibodies diluted in TBST containing 5% non-fat milk at 4 °C, overnight. The dilution of primary antibodies are as follows: *ERα* 1:1000, *ERβ* 1:6000 (Merck, USA) pERK 1:1000, ERK 1:4000, pP38 1:1000, P38 1:2000, cyclin D1 1:500, cyclin B1 1:1000, cyclin E1 1:1000 (Cell signaling, USA), cyclin A 1:500 (BD Biosciences, USA), pERα-Ser118 1:1000, VEGFR2 1:1000 (Santa Cruz Biotechnology, USA) and Beta-actin 1:20,000 (Bio-Rad, USA). Membranes were then washed and incubated with the secondary antibody coupled with HR-peroxidase (1:2000 dilution in blocking buffer containing 0.1% Tween-20) for 2 h. Immunoreactivity was visualized using enhanced chemiluminescence detection (ECL Kit) according to the manufacturer's recommendations (GE Healthcare, UK) and exposed to ECL X-ray film (Amersham, Biosciences, UK).

2.8. Immunofluorescent staining

For immunofluorescent staining, cells were grown in complete medium conditions and cultured until reach to 80%. Then, HuCCA-1 and MCF-7 cells were seeded on cover slips in 24 well plates (1.5×10^4 cells/well) and incubated overnight for cell attachment. Cells were then washed twice with PBS, fixed in 4% paraformaldehyde/PBS for 10 min at 4 °C, subsequently washed and permeabilized with blocking buffer (1× PBS, 1%BSA and 0.2%Triton-X100) for 45 min at room temperature. Cells were subsequently counterstained with anti-rabbit *ERα* clone 60C (Merck, USA) (dilution 1:500 and 1:1000 for HuCCA-1 and MCF-7, respectively) for 1 h. Next, cells were washed and blocked with blocking buffer for 10 min at room temperature. Then cells were incubated at room temperature with the anti-rabbit Rhodamine Red™ IgG secondary antibody (1:400, Jackson ImmunoResearch Laboratories, USA) and counterstained with Hoechst 33342 (nuclei staining; Molecular probes, USA) for 45 min in the dark

at room temperature. Cells were then washed twice with PBS. Finally, cells were mounted with mounting media onto a microscopy slide and images were captured by ImageXpress® High Content Imaging System and analyzed with MetaXpress analysis software (Molecular Devices, USA). The experiments were performed in triplicates with the different cell passages.

2.9. Statistical analysis

All experiments were repeated at least three times with freshly prepared chemicals in each experiment and applied to treat the cell lines in different passages. Data are presented as the means ± standard error (SEM). For cell viability assay, unpaired t-tests were used for comparison between two groups. For protein and mRNA expressions, one-way ANOVA was used for statistic comparison between controls and chemical treatments. For multiple comparisons of data among controls, treatment with and without inhibitors, two-way ANOVA was used to determine statistic comparison. In all cases, *p* value less than 0.05 was considered as statistically significant.

3. Results

3.1. Variation of estrogen receptors (ERs) in cholangiocytes, CCA and breast cancer cells

Previous studies reported that estrogen receptors (ERs) are expressed in different patterns in various types of CCA cell lines (Alvaro et al., 2006). Our initial experiment was performed to determine the expression levels of ERs in MMNK-1, HuCCA-1 and RMCCA-1 cells and compared to that in MCF-7 human breast cancer cells (Fig. 1A). The results showed that HuCCA-1 cells expressed *ERα* but the expression level was much lower than that in MCF-7 cells. Interestingly, *ERα* was not detected in RMCCA-1 and MMNK-1 cells. Meanwhile, *ERβ* was expressed in all cell lines at similar levels. Moreover, the protein level of the ERs was compared among the different culture conditions: CCA cells in complete medium with 10% serum (CM), serum free medium (SF) and charcoal-stripped serum medium (CSS). HuCCA-1 cells displayed an increased *ERα* protein expression in hormone withdrawal medium

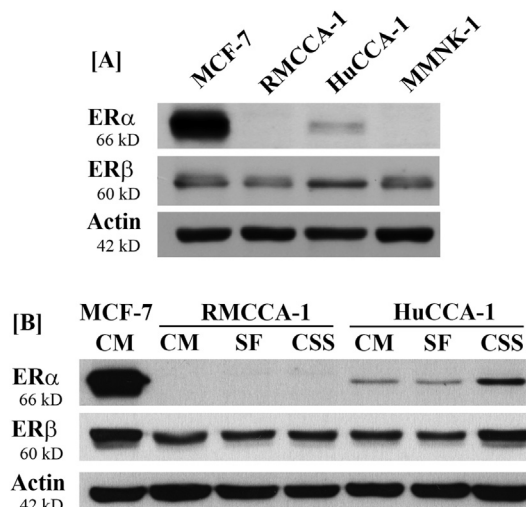


Fig. 1. Protein levels of estrogen receptors (ERs), including *ERα* and *ERβ*, in breast cancer (MCF-7), intrahepatic cholangiocarcinoma (RMCCA-1 and HuCCA-1) and cholangiocyte (MMNK-1) cell lines. [A] The comparison of protein expression among the 4 cell lines cultured in complete medium (10% FBS) conditions (basal level of cell culture condition). [B] Cells were cultured in complete medium with 10% FBS (CM), 2 days of serum free medium (SF) and 5 days of 10% charcoal stripped serum medium (CSS). The hormone-dependent breast cancer cell line, MCF-7, was used as a positive control.

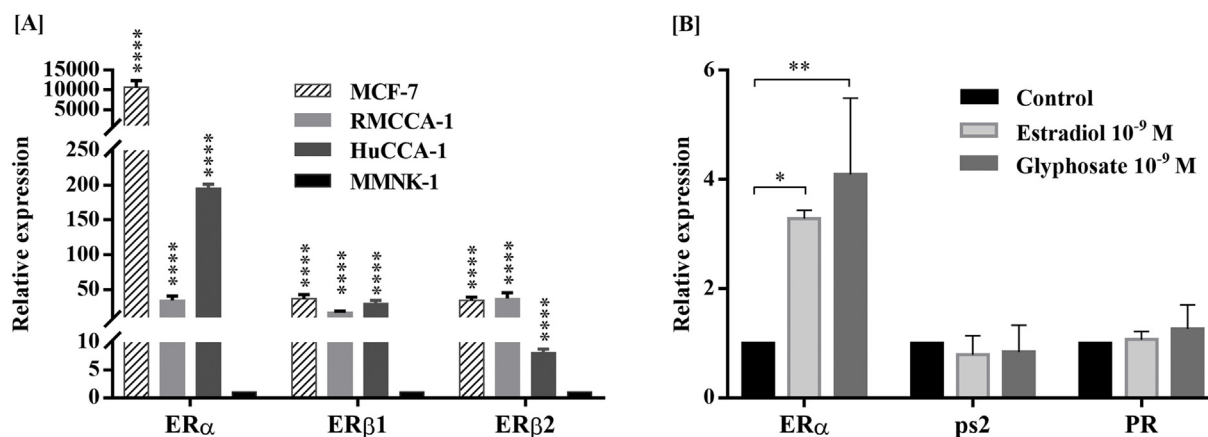


Fig. 2. The relative mRNA levels of genes that involve with estrogen signaling pathway. [A] Relative mRNA levels of *ERα*, *ERβ1* and *ERβ2* in MCF-7, RMCCA-1, HuCCA-1 and MMNK-1 cells in complete medium cell culture (basal level). The cells were grown on 100 mm cell culture plates in complete medium containing 10% FBS with all supplemented chemicals. After 80% cell confluence, RNA was extracted from these cells followed by qRT-PCR analysis. Data are represented as the mean relative expression of controls (MMNK-1 cells) of the three independent assays. [B] Relative mRNA levels of *ERα*, *ps2* and *PR* in HuCCA-1 cells after incubation with E2 and glyphosate at a concentration of 10^{-9} M. HuCCA-1 cells were grown in non-phenol red RPMI containing 10% CSS medium for 5 days prior to treatment. (* = $p \leq 0.05$, ** = $p \leq 0.01$, **** = $p \leq 0.0001$, significantly different as compared to controls).

but not in other conditions (Fig. 1B). These data suggest that the responses to estrogen at the level of ERs expression of MMNK-1, RMCCA-1 and HuCCA-1 cells might be different because these cells had different patterns of ERs expression.

A qRT-PCR was performed to determine the mRNA expression of the *ERα*, *ERβ1*, *ERβ2*, *ps2* and *PR* genes. The results showed that under normal culture conditions (with 10% FBS), all cell types expressed ERs mRNA but in different levels. The mRNA expression of MMNK-1 cells was considered as basal compared to the other cell types. The results showed that the MCF-7 cells had the highest *ERα* mRNA expression level, followed by the HuCCA-1 and RMCCA-1 cells ($10,636 \pm 1,724$, 195 ± 5 and 34 ± 6 relative units, respectively) (Fig. 2A). There was no difference in *ERβ1* mRNA expression in all cell types while HuCCA-1 cells had the lowest level of *ERβ2* mRNA expression when compared among the cancer cells.

3.2. Cell proliferation following estrogen treatment in cholangiocytes and CCA cells

Because different levels of ERs protein and mRNA expression were observed among the MMNK-1, RMCCA-1 and HuCCA-1 cell lines then we hypothesized that the responses of each cell following treatment with estrogen (E2) might be different. To prove the hypothesis, these cells were treated with various concentrations of E2 (10^{-11} , 10^{-9} , 10^{-7} , 10^{-5} M) under estrogen withdrawal conditions. The results showed that only the HuCCA-1 cells underwent significant increase in cell proliferation after E2 treatment, in a dose-dependent manner (Fig. 3A), while no such change was noticeable in the other cell types. These data suggest that *ERα* might be the main target of E2-induced cell proliferation since the HuCCA-1 cells was the only one that expressed *ERα* while none of the other tested cells.

3.3. Glyphosate and E2-induced HuCCA-1 cell proliferation

Though, previous *in vitro* studies have demonstrated that estrogen can induce CCA cells proliferation (Alvaro et al., 2006; Mancino et al., 2009) and glyphosate can induce breast cancer cell growth via the estrogen receptor (Thongprakaisang et al., 2013). Therefore, it becomes crucial to figure out that glyphosate might induce CCA cell proliferation via estrogen signaling. To achieve this, a cell viability assay of *ERα*-expressing CCA cell line, HuCCA-1, was performed under estrogen-withdrawal conditions. To determine the proper conditions for treatment we performed two different conditions, 2 days SF and 5 days CSS

treatments prior to carrying out the Prestoblu cell viability assay. The results showed that both E2 and glyphosate exerted the proliferative effects in these cells only in the CSS condition (Fig. 3B). Thus, we used CSS conditions in all further subsequent experiments. In addition, HuCCA-1 cells were treated with various concentrations of glyphosate (10^{-15} to 25×10^{-3} M) and E2 (10^{-15} to 10^{-5} M). After 48 h of incubation, cell viability analysis was carried out through the Prestoblu assay (Fig. 3C). The results showed that the low concentrations of glyphosate and E2 significantly induced HuCCA-1 cell proliferation. The lowest concentrations of glyphosate and E2 that induced HuCCA-1 cell proliferation were 10^{-13} and 10^{-11} M, respectively. Glyphosate at low concentration (10^{-13} – 10^{-7} M) exhibited a higher proliferative effect than E2 in HuCCA-1 cells. However, this induction of cell proliferation was decreased at higher concentrations (10^{-5} M) of glyphosate but not E2 treatment indicated cytotoxicity of glyphosate at the high concentrations (10^{-3} to 25×10^{-3} M). The MTT assay provided similar results to Prestoblu assay (data not shown). These data suggested that glyphosate might induce HuCCA-1 cell proliferation via the estrogen signaling pathway.

3.4. Glyphosate and E2 induce *ERα* mRNA expression but not in *ERE* responsive genes

The estrogen signaling pathway was previously categorized into genomic and non-genomic signaling. The genomic or classical signaling pathway of estrogen was analyzed by qRT-PCR. The mRNA expression of genes under genomic signaling included: *ERα*, *ps2* and *PR*, were determined. The results showed that glyphosate and E2 induced only *ERα* mRNA expression, but did not result in significant changed in *ps2* and *PR* mRNA levels (Fig. 2B). These data suggested that glyphosate and E2 might not have effects mediated through the genomic/classical of estrogen signaling pathway.

3.5. Glyphosate and E2 effects on various signaling protein expressions in CCA cells

We demonstrated that the induction of cell proliferation in HuCCA-1 cells occurred through estrogen signaling but not in the genomic or classical signaling pathway. As a next step, the expressions of signaling proteins related to the non-genomic signaling pathway of estrogen were analyzed. The results demonstrated that glyphosate and E2 altered the expression levels of several signaling proteins including *ERα*, *ERβ*, VEGFR2, phosphorylated ERK (pERK) and ERK (Fig. 4A and B). At 48 h

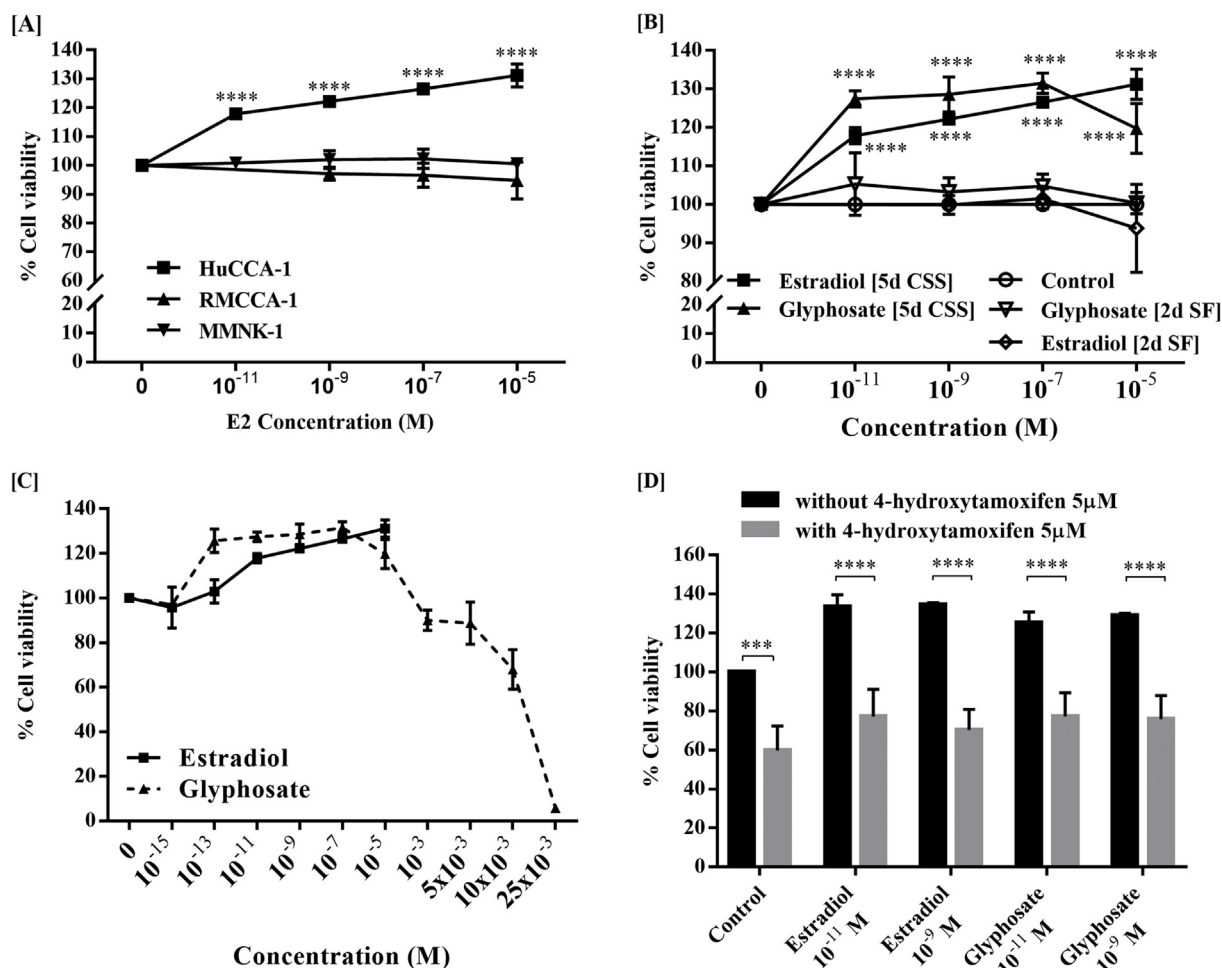


Fig. 3. The effects of estradiol (E2) and glyphosate on cell viability. [A] Effects of E2 on RMCCA-1, HuCCA-1 and MMNK-1 cell viability in hormone withdrawal medium condition. The cells were grown in 10% charcoal stripped serum medium (CSS) for 5 days and treated with 10^{-11} to 10^{-5} M of E2 for 48 h in 5% CSS medium. [B] HuCCA-1 cells were cultured in 2 different conditions prior to treatment: serum free medium for 2 days (2d SF) and 10% charcoal stripped serum medium for 5 days (5d CSS) followed by treatment with 10^{-11} to 10^{-5} M of estradiol or glyphosate for 48 h. [C] HuCCA-1 cells were treated with 10^{-15} to 10^{-5} M of estradiol or 10^{-15} to 2.5×10^{-2} M of glyphosate for 48 h in 5% CSS medium. [D] An ER antagonist inhibited the proliferative effects of E2 and glyphosate. HuCCA-1 cells were treated with 10^{-11} M and 10^{-9} M of E2 and glyphosate and co-incubated with 5 μ M 4-hydroxytamoxifen (Tamoxifen). Cell viability was measured by the Prestobluo viability assay. Each data point represents the mean $\pm$ SE of 3 independent experiments and expressed relative to controls ($n = 3$, *** = $p \leq 0.001$, **** = $p \leq 0.0001$; [A], [B] and [C] significantly different as compared to controls, [D] significantly different as compared between with and without antagonist).

of exposure, all tested concentrations (10^{-11} , 10^{-9} and 10^{-7} M) of glyphosate increased the levels of ER α while only 10^{-11} M E2 induced ER α expression. The level of ER β did not change following E2 or glyphosate exposure. The ratio of pERK/ERK was also up-regulated after treatment with E2 and glyphosate. We also demonstrated that E2 and glyphosate increased the level of VEGFR2 expression in HuCCA-1 cells. Previous study demonstrated that E2 induced the expression of IGF1-R protein levels in HuH-28 CCA cells (Alvaro et al., 2006), but we could not detect the change in IGF-1R protein expression (data not shown). These results suggest that glyphosate altered the expression of ER α and other signaling proteins related to the non-genomic signaling pathway of estrogen especially ERK1/2 pathway.

3.6. Cell cycle distribution effects after treatment with E2 and glyphosate

Since E2 and glyphosate induced HuCCA-1 cell proliferation, next we wanted to investigate whether or not these chemicals will alter cell cycle distribution. After 48 h of E2-depleted treatment with E2 and glyphosate, cell cycle were measured by using flow cytometry and PI-staining, the results demonstrated that both chemicals in all tested concentrations increased cells in the S-phase (Fig. 5A). Moreover, 10^{-11} M E2 and 10^{-9} M glyphosate also decreased G2/M phase cells.

3.7. Glyphosate and E2-induced the levels of cyclin signaling protein expressions

Our data demonstrated that glyphosate and E2 increased S-phase cell cycle distribution. Next, we analyzed the expression of cyclin signaling proteins involved in cell cycle regulation. The results showed that expressions of all cyclins were increased after 48 h of exposure, especially cyclin D1 and cyclin A, which are involved in driving the cells to the S-phase (Fig. 6A and B).

3.8. The proliferative effects of glyphosate are mediated via estrogen receptors

From the proliferative effect of glyphosate observed in HuCCA-1 cells in the absence of E2, it was hypothesized that ER signaling may be involved in this glyphosate-induced CCA cell proliferation. HuCCA-1 cells were further studied using an ER antagonist, 5 μ M 4-hydroxytamoxifen, which was previously reported to inhibit growth of human CCA cells (Sampson et al., 1997). The results showed that 4-hydroxytamoxifen inhibited the proliferative effects of glyphosate and E2 (Fig. 3D). Furthermore, 4-hydroxytamoxifen inhibited the induction of S-phase cell populations by glyphosate and E2 (Fig. 5B). We also

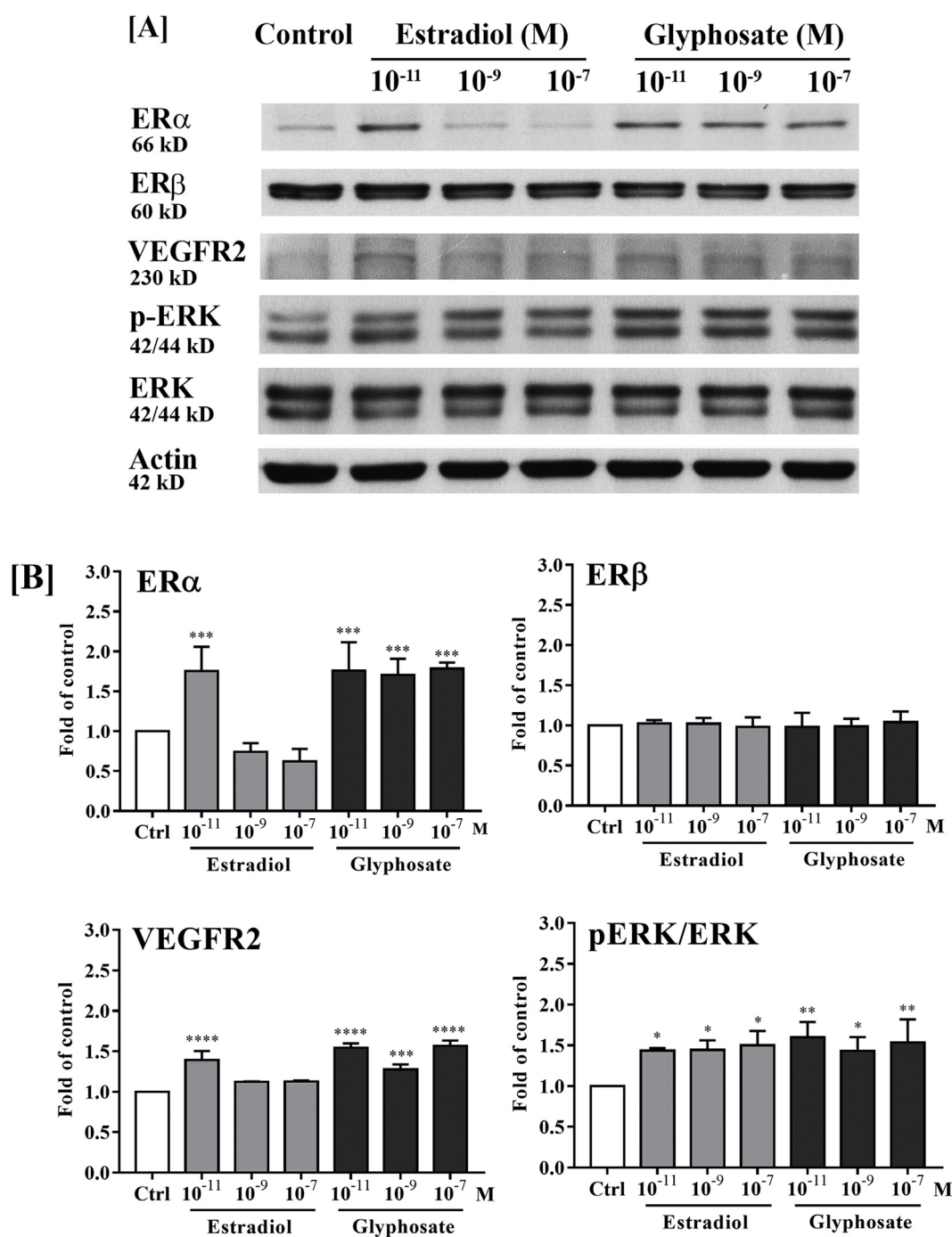


Fig. 4. Effects of estradiol and glyphosate on the level of protein expression in E2-withdrawal HuCCA-1 cell culture. [A] 48 h incubation time resulted in specific bands for ERα, ERβ, VEGFR2, phosphorylated-ERK (pERK), total ERK and β-actin (representative sample from one experiment). Optical densities of specific bands were determined from western blots and each band was normalized to the β-actin band. [B] The normalized mean of three replications $\pm$ SE optical density values are shown in the histogram for ERα, ERβ, VEGFR2, p-ERK/ERK ($n = 3$, * = $p \leq 0.05$, ** = $p \leq 0.01$, *** = $p \leq 0.001$, **** = $p \leq 0.0001$, significantly different as compared to controls).

performed Western immunoblotting to determine the expression of several signaling proteins after treatment with glyphosate, E2 and 4-hydroxytamoxifen. The results showed that after co-treatment the chemicals with ER antagonist, 4-hydroxytamoxifen, all signaling proteins that involved with cell proliferation and non-genomic signaling of estrogen, including pERα(Ser118), ERα, VEGFR2, pERK/ERK, PI3K (p85), cyclin A and PCNA, had decreased levels of expression compared to chemicals treatment alone (Fig. 7A and B). These data suggest that the effects of glyphosate occurred via estrogen signaling in the HuCCA-1 cells.

3.9. Glyphosate effects on HuCCA-1 were partly mediated through the MEK signaling pathway

We demonstrated that glyphosate induced CCA cell proliferation and altered expression levels of several signaling proteins, such as pERK, cyclin D1 and cyclin A. Next, we used a MEK1/2 inhibitor ($10 \mu\text{M}$ U0126) to determine whether this signaling pathway is related to glyphosate effects or not. The results showed that U0126 decreased the expression level of several proteins, including pERK, pP38 and cyclin D1 (Fig. 8A and B). The reduction of PCNA suggested that U0126

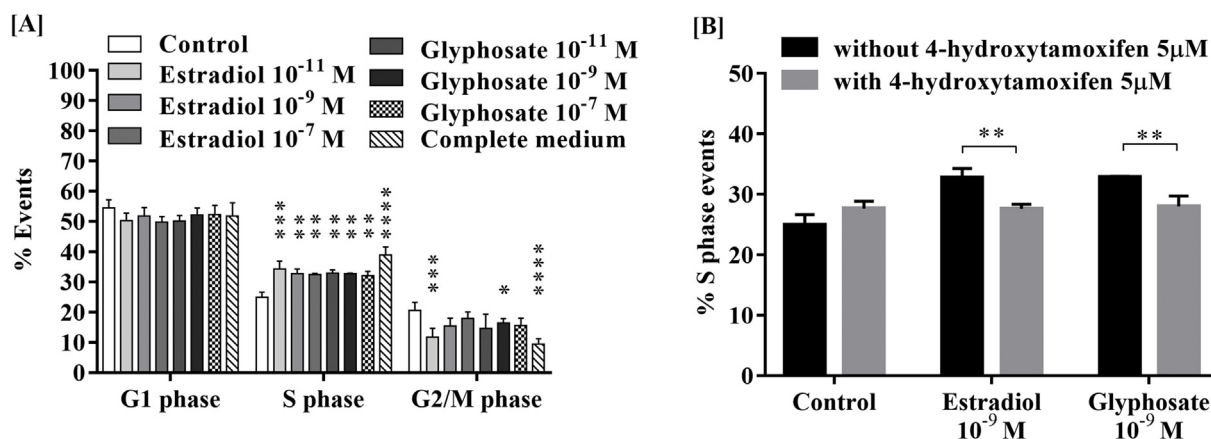


Fig. 5. The demonstration of a concentration-effect on cell cycle by estradiol and glyphosate in HuCCA-1 cells. [A] Cells were treated with varying concentrations (10^{-11} to 10^{-5} M) of estradiol and glyphosate. Cell cycle distributions were compared in hormone withdrawal medium (CSS medium) by using PI staining flow cytometry at 48 h. [B] An ER antagonist inhibited S-phase induction of E2 and glyphosate. HuCCA-1 cells were treated with 10^{-11} M and 10^{-9} M of estradiol and glyphosate and co-incubated with 5 μ M 4-hydroxytamoxifen (Tamoxifen). (n = 3, * = $p \leq 0.05$, ** = $p \leq 0.01$, *** = $p \leq 0.001$, **** = $p \leq 0.0001$; [A] significantly different as compared to controls, [B] significantly different as compared between with and without antagonist co-treatment).

inhibited the proliferation effect of both E2 and glyphosate. These data indicated that the effects of glyphosate were partly mediated through the MEK signaling pathway.

3.10. Different localization of ER α between HuCCA-1 and MCF-7

To determine ER α localization in CCA and breast cancer cells, HuCCA-1 cells and MCF-7 were grown in complete medium conditions on coverslip and examined by immunofluorescent staining technique. The result showed a punctate ER α expression that was mainly localized in the cytoplasmic compartment up to $89.1 \pm 7.5\%$ in HuCCA-1 cells, whereas $88.7 \pm 2.3\%$ was primarily localized in the cell nucleus in MCF-7 cells (Fig. 9). The cytoplasmic marker, α -tubulin antibody was also co-stained to ensure the subcellular localization of the ER α in the cytoplasm of the cell (data not shown). Our results indicated that the estrogen signaling pathway of HuCCA-1 cells may be related with non-genomic action, while estrogen signaling pathway of MCF-7 cells is mainly genomic mechanism.

4. Discussion

This is the first study that demonstrates the effects of glyphosate on cholangiocarcinoma cells, and provides a better understanding of the possible mechanisms related to its effects. The low concentrations of glyphosate tested in this study are relevant to the range of levels that have been reported in the environment and detected in humans (Acquavella et al., 2004; Bai and Ogbourne, 2016; Kongtip et al., 2017).

Previous studies have reported that E2 induces cell proliferation in an ER α expressing CCA, HuH28 cell line (Alvaro et al., 2006). It was reported that E2 also induced the expression levels of several proteins, including ER α , VEGF, VEGFR, IGF-1, and IGF-1R, as well as downstream signaling proteins associated to cell proliferation. In this study, we found that 10^{-11} to 10^{-5} M of glyphosate and E2 induced cell proliferation in HuCCA-1 cells under estrogen withdrawal conditions and the inductive effects were inhibited by ER antagonists. This study also demonstrated the results of various doses of glyphosate that covered no effect (10^{-15} M) and toxicity (10^{-3} to 25×10^{-3} M) in HuCCA-1 cells. These results demonstrated that low concentration of glyphosate induced bile duct cancer cell growth via the estrogen receptor which is similar to the previous report in breast cancer cell line (Thongprakisang et al., 2013). A recent report from other researchers showed that only high concentration of glyphosate can induce cell proliferation in breast cancer cells (Mesnage et al., 2017). The different results of these two studies may reflect the different estrogen

withdrawal durations. This study demonstrated that the estrogen withdrawal duration had an effect on analysis of exogenous estrogen effects. We showed that glyphosate and E2 did not induce cell proliferation of HuCCA-1 cells in 2-days SF condition, which was comparable to 2-days CSS condition. Interestingly, 5-days in CSS prior to glyphosate or E2 treatment resulted in cell proliferation induction in the HuCCA-1 cells (Fig. 3B). The ER α protein expression in different culture conditions supported this data because 5 days CSS culture condition displayed higher ER α protein expression than other culture conditions (Fig. 1B). The higher levels of estrogen receptor expression might increase the response to weak binding chemicals, such as glyphosate in this study. The induction of ER α mRNA and protein levels suggests that they may have a role in CCA progression; however, we did not detect a change in ER β in the HuCCA-1 cells after treatment with E2 and glyphosate. This result is similar to that from a previous study which showed that E2 can induce ER α expression but lower response in ER β expression (Alvaro et al., 2006).

Glyphosate-induced cell proliferation effects were also supported by cell cycle analysis and confirmed that the responses occurred via the estrogen receptor because these effects were inhibited by an ER antagonist (4-hydroxytamoxifen). In addition, the preliminary unpublished data in our study showed that ICI 182,780 alone at low concentrations (10^{-9} – 10^{-8} M) act as agonist to estrogen receptor in HuCCA-1 cell line, while 4-hydroxytamoxifen did not. Previously, ICI 182,780 has been reported that it induces cell proliferation and acts as ER α agonist in the other cells beside breast cancer cells (Hanazono et al., 1998; Zhao et al., 2006). Therefore, we used only 4-hydroxytamoxifen in this study. There are some studies that support the observation that induction of cell proliferation is concomitant with an increase in S-phase cell populations (Amonyingcharoen et al., 2015; Bertoli et al., 2013). Two different cyclin proteins, cyclin A and cyclin D1, have been reported as activators of the estrogen receptor (Neuman et al., 1997; Trowbridge et al., 1997). In this study, we found that E2 and glyphosate exposure leads to an increase in the protein levels of cyclin D1, and cyclin A, as well as other cyclins involved in cell cycle regulation. In our study, glyphosate induces a higher proliferative effect than E2 in HuCCA-1 cells with a significant difference at low concentration, 10^{-13} – 10^{-9} M. At higher concentration (10^{-7} – 10^{-5} M), there is no significant difference between these two chemicals. At present, we can't exactly explain this higher proliferative action of glyphosate. However, in a recent study of Mesnage and colleague, they reported the molecular dynamics simulation of glyphosate and estradiol binding to human estrogen receptor alpha (ER alpha) and they concluded that glyphosate is unlikely to activate ER alpha in a similar

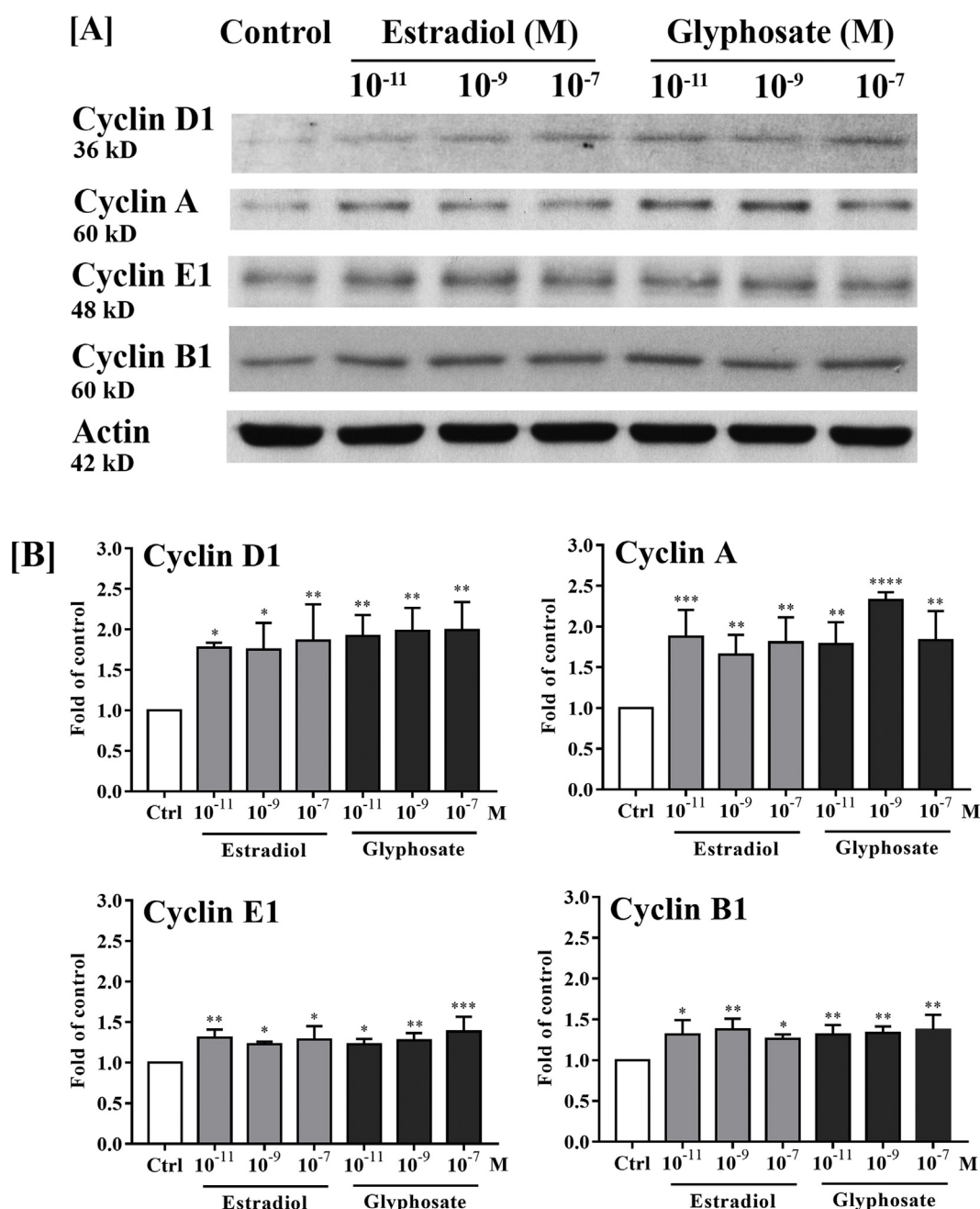


Fig. 6. The effects of estradiol and glyphosate on the level of cell cycle regulated protein expressions in E2-withdrawal HuCCA-1 cell culture. [A] 48 h incubation time resulted in specific bands for cyclin D1, cyclin A, cyclin E1, cyclin B1 and β -actin (representative sample from one experiment). Optical densities of specific bands were determined from western blot and each band was normalized to the β -actin band. [B] The normalized mean of three replications $\pm$ SE optical density values are shown in the histogram for ER α , ER β , VEGFR2, p-ERK/ERK (n = 3, * = $p \leq 0.05$, ** = $p \leq 0.01$, *** = $p \leq 0.001$, **** = $p \leq 0.0001$, significantly different as compared to controls).

manner to the natural hormone (Mesnage et al., 2017). Furthermore, we observed the different effects of glyphosate and E2 on the ER α expression. All tested concentrations of glyphosate increased ER α expression whereas E2 only at 10^{-11} M increased ER α expression (Fig. 4). Estrogen or estradiol which is natural hormone has negative feedback inhibition mechanism after binding and activate estrogen receptor. Drabovich and co-workers showed that there is a network of transcription factors with a negative feedback loop (ER α -EGR3-NAB2) regulated the estrogen activation in breast cancer cells (Drabovich et al., 2016). Glyphosate which is a chemical mimic natural hormone may not be inhibited by the activation of the negative feedback as natural hormones do. However, this hypothesis need further study.

It should be noted that not only glyphosate but E2 at the same concentration range (10^{-11} - 10^{-7} M) also showed similar effects on the expression of the cell cycle regulating proteins. Furthermore, we could not see a significant difference among this concentration range of glyphosate and E2 on other tested endpoints including cell proliferation, ERK activation and S-phase cell cycle. In contrast to our previous study in breast cancer cell lines, glyphosate at the same concentration showed lower proliferative induction than E2 (Thongprakaisang et al., 2013). Since the maximum saturation of the effects of glyphosate and E2 on cell proliferation in HuCCA-1 were observed at low concentrations (10^{-11} - 10^{-9} M) together with the maximum proliferative response of these two compounds were relatively low (< 30% of control), we

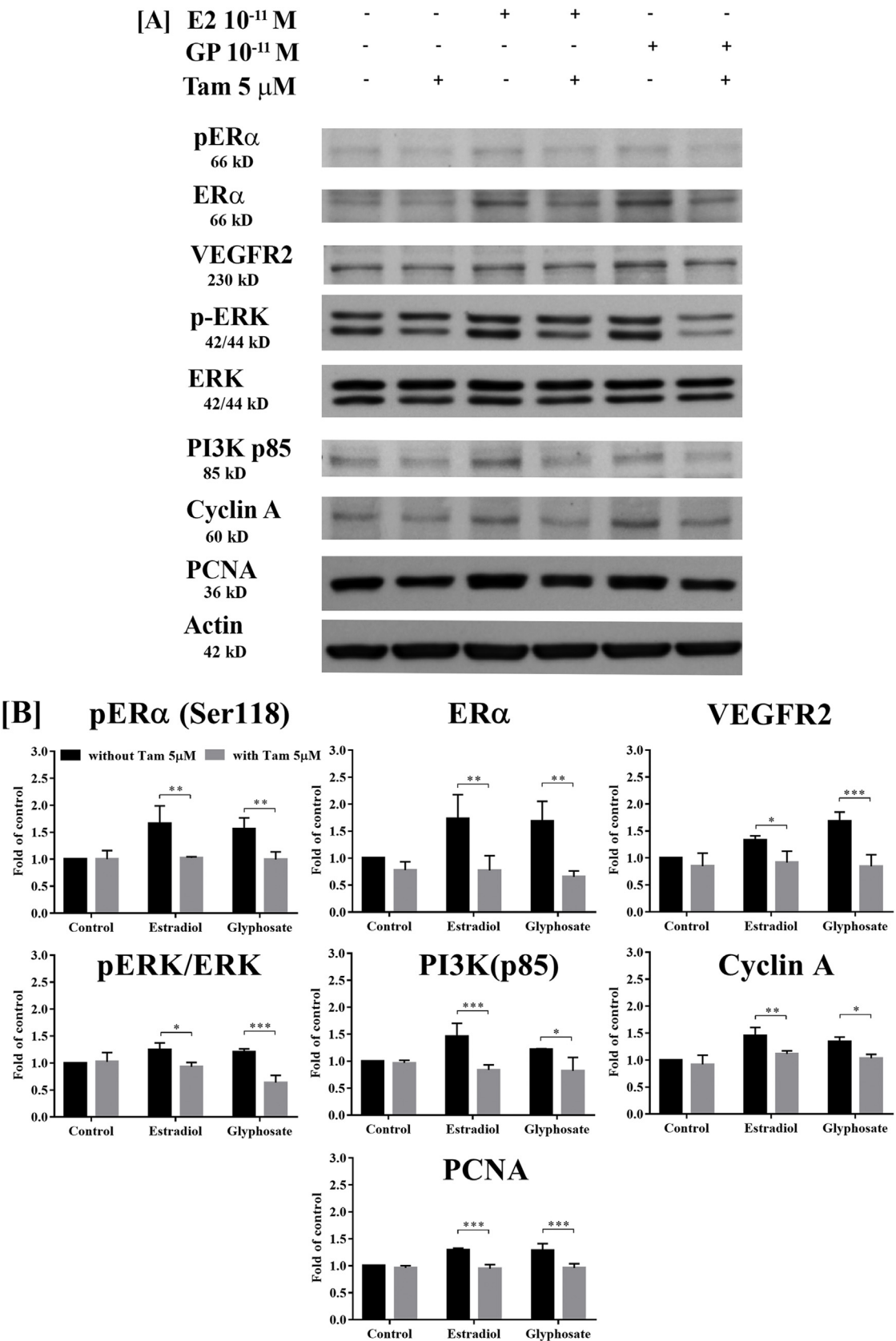


Fig. 7. Effects of an ER antagonist (4-hydroxytamoxifen, Tam) on the levels of protein expressions after co-treatment with E2 and glyphosate (10⁻¹¹ M) in E2-withdrawal HuCCA-1 cell culture. [A] 48 h incubation time resulted in specific bands for ERα, VEGFR2, phosphorylated-ERK (pERK), total ERK, PI3K(p85), cyclin A, PCNA and β-actin (representative sample from one experiment). Optical densities of the specific bands were determined from western blot and each band was normalized to the β-actin band. [B] The normalized mean of three replications ± SE optical density values are shown in the histogram for ERα, ERβ, VEGFR2 and p-ERK/ERK (n = 3, * = p ≤ 0.05, ** = p ≤ 0.01, *** = p ≤ 0.001, **** = p < 0.0001, significantly different as compared to controls).

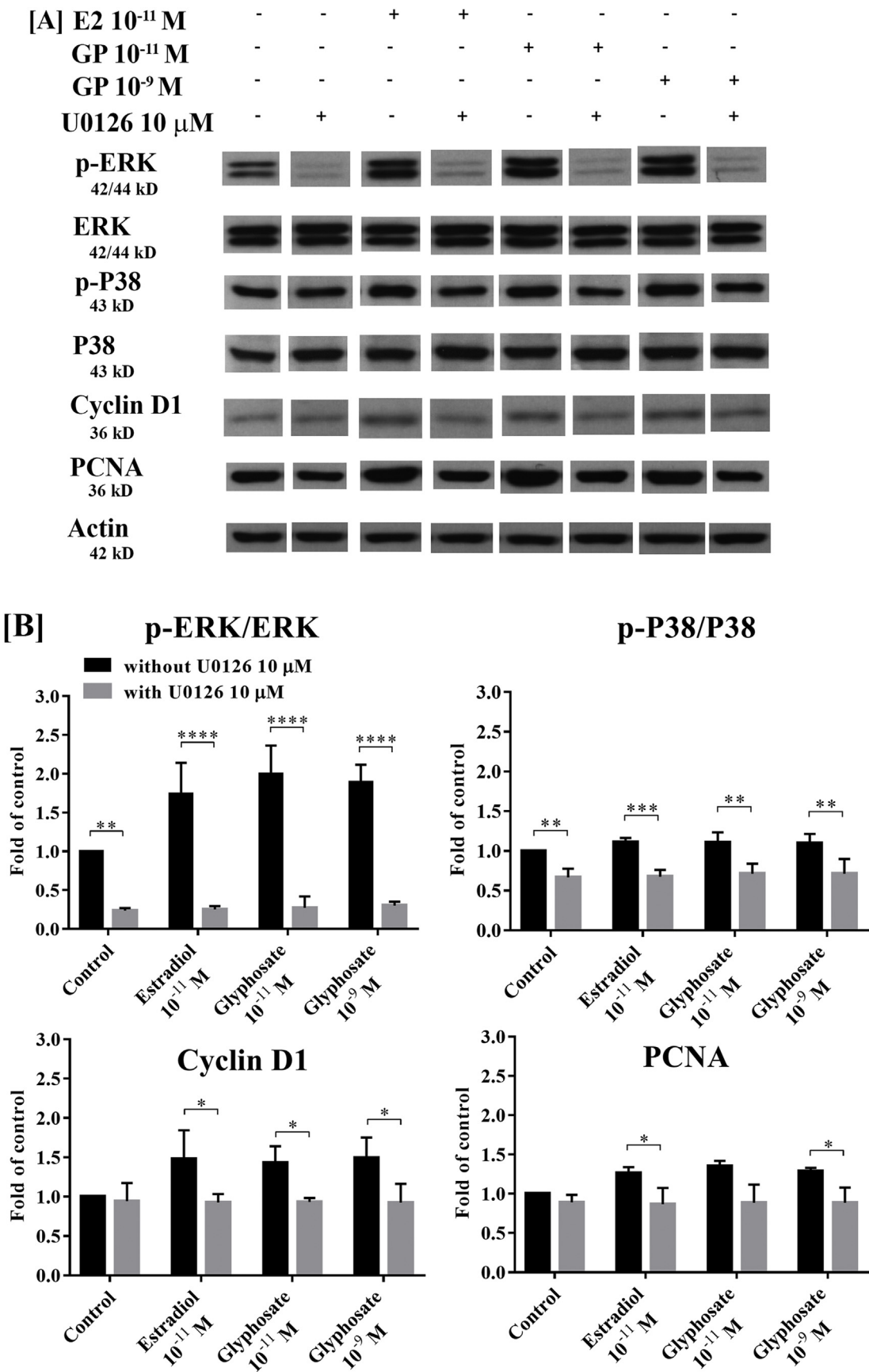


Fig. 8. The effects of glyphosate on signaling protein expression. E2-withdrawal HuCCA-1 cells were used. [A] 10 μ M of MEK inhibitor (U0126) was applied for co-incubation with HuCCA-1 cells. 48 h incubation resulted in specific bands for p-ERK, ERK, p-P38, P38, cyclin D1, PCNA and β -actin (representative sample from one experiment). Optical densities of specific bands were determined from western blot and each band was normalized to the β -actin band. [B] The normalized mean of three replications $\pm$ SE optical density values are shown in the histogram for expression of different proteins ($n = 3$, * = $p \leq 0.05$, ** = $p \leq 0.01$, *** = $p \leq 0.001$, **** = $p < 0.0001$, significantly different as compared between with and without inhibitor co-treatment).

hypothesize there is a rapid estrogen receptor saturation and low sensitivity of non-genomic estrogen receptor signaling to induce cholangiocarcinoma HuCCA-1 cell growth. It is also postulated that non-genomic estrogen receptor signaling occurred faster than genomic estrogen receptor signaling pathway. Therefore, the time point that we determined the effects are occurred at the near maximal responses. However, further study to prove this hypothesis is really needed.

The molecular mechanisms of estrogen activity can be categorized into two main pathways: genomic and non-genomic signaling pathways. In the genomic or classical pathway, E2 binding triggers conformational changes in the receptor leading to receptor dimerization. Active ER dimer then binds directly or indirectly to genes containing estrogen response elements (ERE). Various co-activators of transcription factors may also be recruited in this step. Transcription of genes, such as *ps2* and *progesterone receptor*; *PR*, is stimulated, resulting in the activation of several cell functions, such as cell proliferation and cell survival (Heldring et al., 2007; Nilsson et al., 2001; Zhang and Trudeau, 2006). Many signaling molecules are rapidly activated by the non-genomic effects of E2 such as epidermal growth factor receptor (EGFR), insulin-like growth factor I receptor (IGF-IR), Ras/Raf-1, mitogen activating protein kinase (MAPK), Akt and protein kinase C (Cheskis, 2004; Yee and Lee, 2000). In this study, we found that estrogenic activity of glyphosate and E2 on HuCCA-1 cells are mainly associated with non-genomic estrogen signaling because we did not find the alteration of gene expressions that were regulated through ERE activation such as *ps2* and *PR*. Rather, we demonstrated that glyphosate and E2 induced the expression levels of non-genomic signaling proteins, such as pERK, VEGFR2. Previous studies showed that estradiol induced the increase of

IGF-1, IGF-1R, VEGF and VEGFR protein levels, and that was inhibited by ER- or IGF-1R-specific antagonists (Alvaro et al., 2006; Mancino et al., 2009). However, we found that E2 and glyphosate induced the protein expression of ER α , and VEGFR2, but not IGF-1R. These different results may be due to different CCA cell lines that may have different responsive mechanisms to chemicals. Moreover, the inductions of those signaling proteins were diminished by an ER antagonist. We also report here that the effects of glyphosate are mediated partly through MEK signaling pathway, as observed from the inhibitory effects of the MEK1/2 inhibitor (U0126).

There are limit data of ER α protein localization in cholangiocarcinoma. We first examined ER α protein expression and localization in HuCCA-1 compared to MCF-7 using immunofluorescent staining. Previously, several studies showed the localization of breast cancer cell lines or patient samples by different techniques, including cell fractionated western immunoblotting, *in situ* proximity ligation assay (PLA), immuno-electron microscopy, immunohistochemistry and immunofluorescent staining (Iwabuchi et al., 2017; Kocanova et al., 2010; Li et al., 2015; Welsh et al., 2012). Li and colleague showed that there are two types of ER α localization in breast cancer cells from patients by immunohistochemical staining; in the cell nucleus and/or the cell membrane (Li et al., 2015). Iwabuchi and colleague presented the dimerization and localization of ER α on MCF-7 and T-47D cells by using double-stained ER α proteins using two different anti-ER α antibodies as well as *in situ* Proximity Ligation Assay (PLA) (Iwabuchi et al., 2017). They showed that ER α signals were detected in the nuclei of both cells. Moreover, cytoplasmic expression of ER α was low incident in cell lines and clinical samples of breast cancer using quantitative

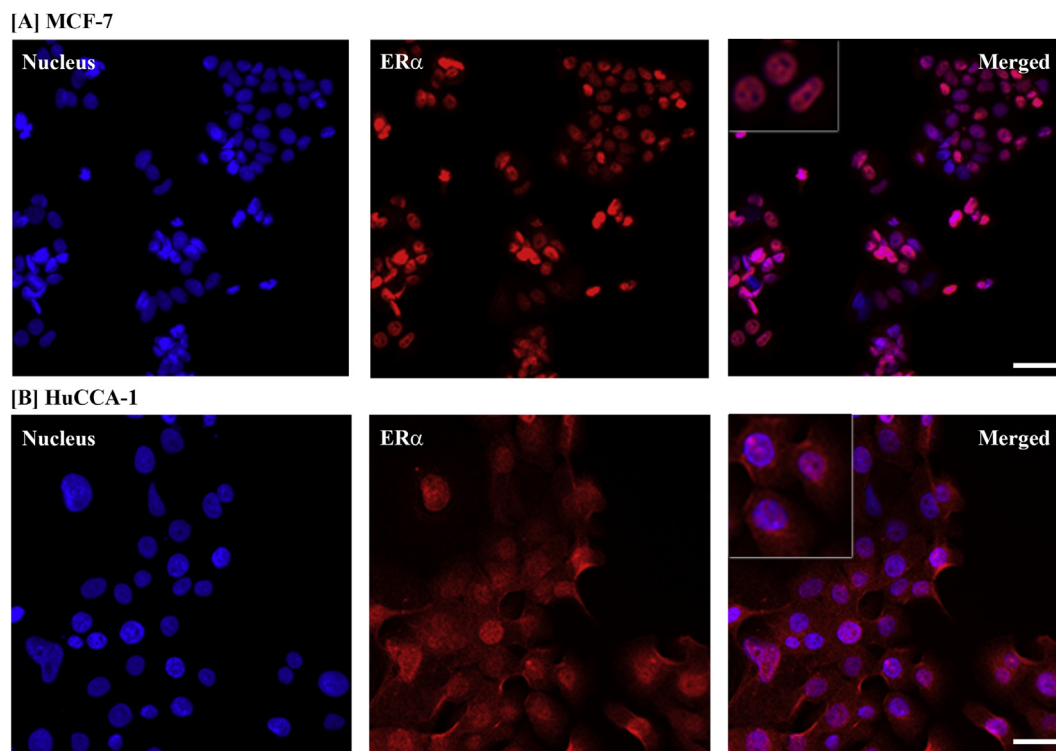


Fig. 9. Detection of cytoplasmic and nuclear ER α expression by immunofluorescent staining. ER α expression and localization in MCF-7 [A] and HuCCA-1 [B] cells, (Rhodamine red, red). Cell nuclei were stained by Hoechst 33342 (blue). Images were captured at 40 $\times$ magnification. Scale bar was at 20 μ m. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

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